Analyzing Credit Risk Using Classification Models

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1.0 Introduction

1.1 Project Description In this project we used machine learning to analyze a loan default dataset and find the strongest predictors of borrowers defaulting on their loans. We trained several logistic regression models and a random forest to predict whether a loan applicant will default, and compared them on the same held-out test set.

1.2 Background When creditors (lenders such as banks) decide who to lend money to, they must do so cautiously. Many borrowers are not able to pay back money that is lent to them with interest, and each default is a loss for the lender.

To reduce those losses, lenders need to estimate how likely each applicant is to repay, using data from past loans.

The goal of this analysis was to use a dataset of borrowers' financial data and loan outcomes to find out which factors lenders should take into account when deciding who to lend money to, and how well those factors predict defaults.

2.0 Data Description

The dataset we used is the Credit Risk Dataset from Kaggle. It contains 32,581 rows of 12 variables. It is a simulation of credit bureau data made for educational purposes, so our results describe this dataset rather than real lending. The variables are:

person_age - numeric: borrower's age in years

person_income - numeric: borrower's annual income in dollars

person_home_ownership - categorical: borrower's home ownership status

person_emp_length - numeric: borrower's length of most recent employment in years

loan_intent - categorical: borrower's reason for taking out the loan

loan_grade - categorical: the lender's risk grade for the loan (A to G)

loan_amnt - numeric: loan amount in dollars

loan_int_rate - numeric: loan interest rate

loan_status - categorical: 0 if the borrower did not default, 1 if the borrower defaulted

loan_percent_income - numeric: loan_amnt / person_income

cb_person_default_on_file - categorical: Y if the borrower has a past default on file, N otherwise

cb_person_cred_hist_length - numeric: length of the borrower's credit history in years

We started with 32,581 rows of data. First, we removed every row with a missing value, which left 28,638 observations.

Age, employment length, and income all had extreme outliers, including ages of up to 144 years and employment lengths of up to 123 years. The box plots below show them.

Fig 2.1: Box Plot for Income

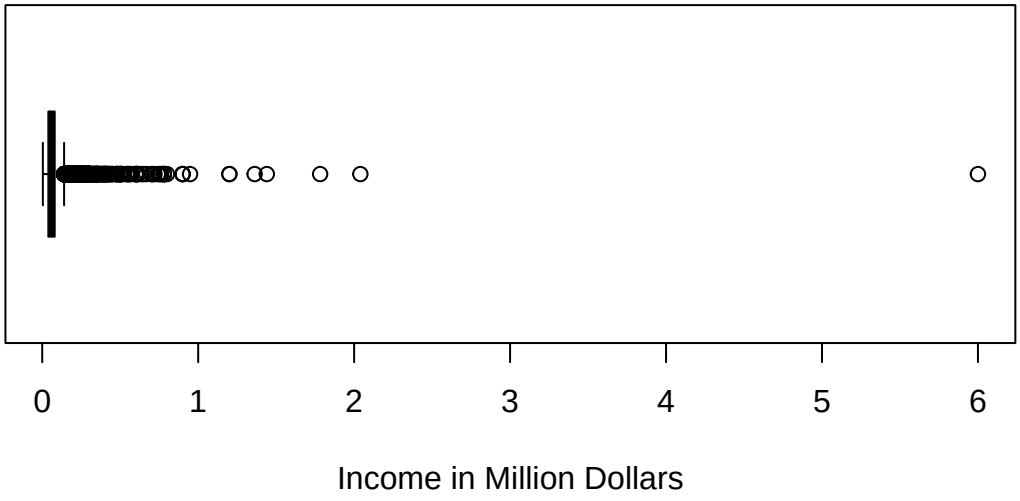
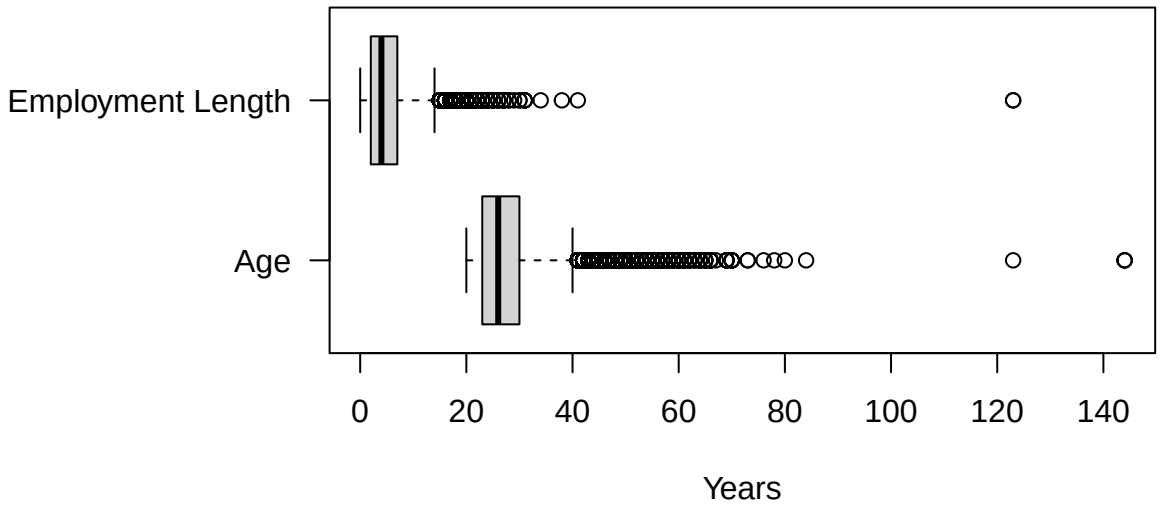


Fig 2.2: Box Plot for Age and Employment Length

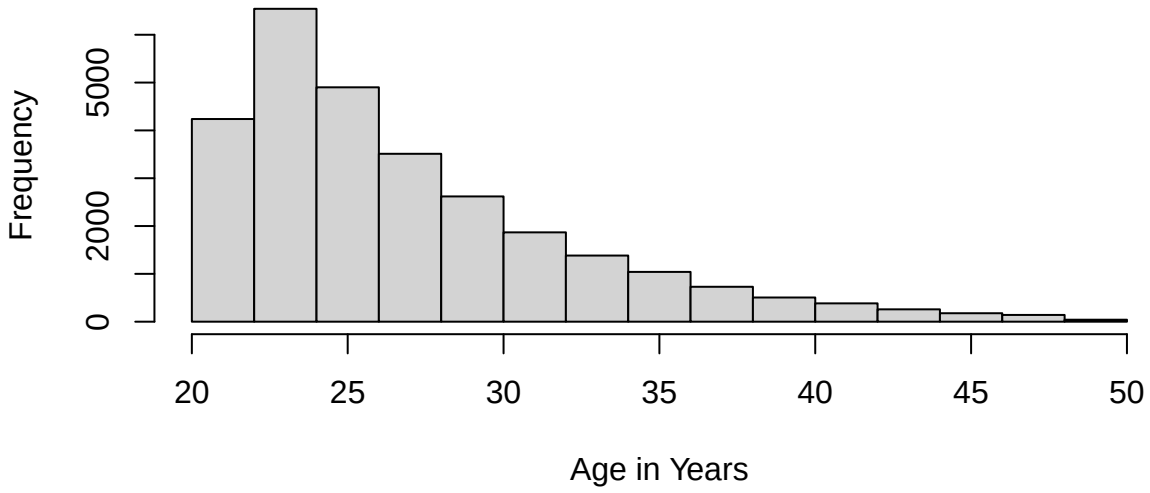


A common rule for removing outliers is to drop any value more than 1.5 interquartile ranges above the third quartile or below the first quartile. Applied to every numeric variable, this rule would remove 17.1% of our rows, and many of those values are realistic: a 45-year-old borrower or a $120,000 income is unusual in this dataset but not an error. Instead, we inspected each variable with outliers and set a cutoff where values became unrealistic or too sparse to draw conclusions from.

For age, some values were impossible (Fig 2.2), but most of the older ages were realistic; 1,236 borrowers were in their 40s. Borrowers aged 50 and older were too few to model reliably, so we removed them.

28,340 observations remain.

Fig 2.3: Histogram of Borrower Ages



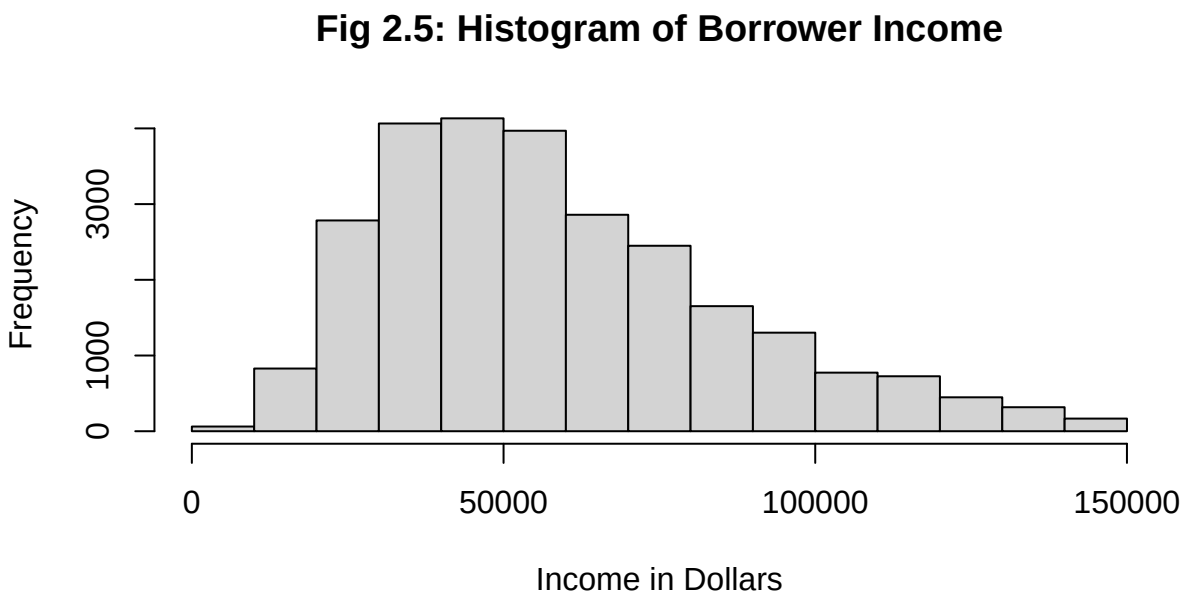
Employment length also has a lot of natural variation, so we did not remove every outlier. The extreme values are clear in Fig 2.2. We removed employment lengths of 15 years or more, where observations became sparse compared with shorter lengths. Figure 2.4 shows the remaining distribution.

27,599 observations remain.



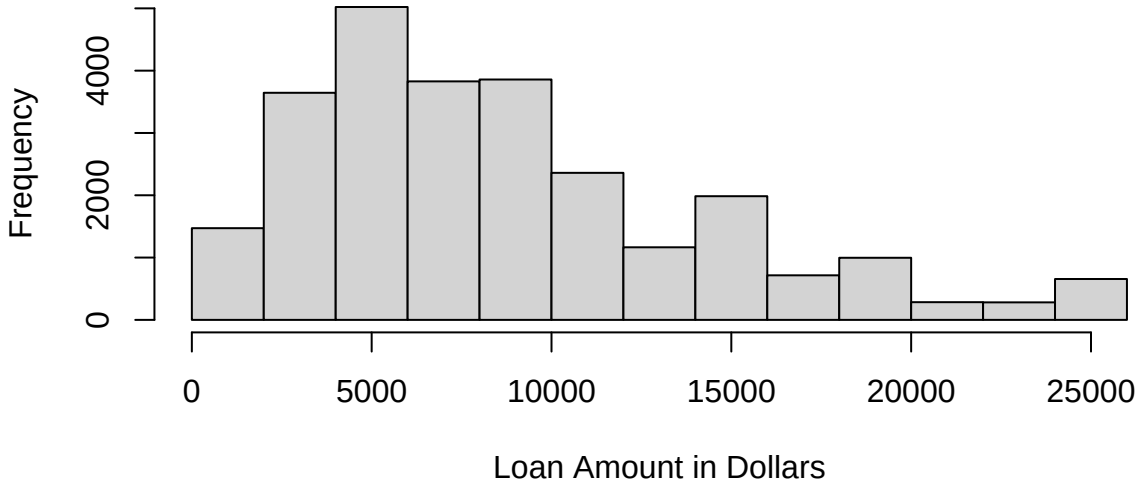
Income varies the most of any variable. Its distribution is right-skewed: half of borrowers earn between $39,000 and $79,000, and a long tail extends past $1,000,000 (Fig 2.1). Observations become sparse above $150,000, so we removed incomes of $150,000 or more. Figure 2.5 shows the remaining distribution.

26,536 observations remain. The median income is now $54,000.



Like most of our numeric variables, loan amount is right-skewed. The number of loans drops sharply above $25,000, so we removed loans larger than $25,000. We kept loans of exactly $25,000, which are common in the data (567 loans).

Fig 2.6: Histogram of Loan Amount



After cleaning, 26,268 observations remain; 6,313 were removed in total. Of these, 5,805 borrowers defaulted (22.1%) and 20,463 did not.

3.0 Analysis

3.1 Training and test sets We split our data once into a training set and a test set: 75% of the data (19,702 observations) for training and the remaining 25% (6,566 observations) for testing. Every model below is evaluated on this same test set, which keeps the real default rate of 22.1% and is never used for training or for choosing predictors.

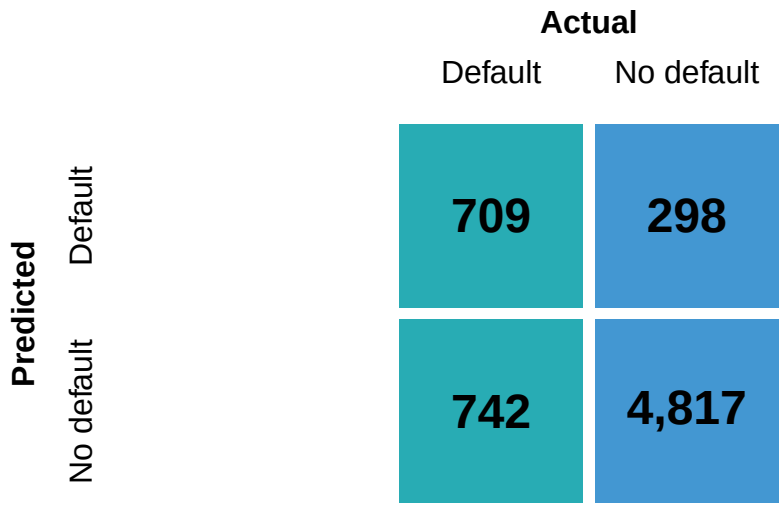
We treat "default" as the positive class. The measures we report are:

- **Accuracy:** the share of all predictions that are correct.
- **Default recall (sensitivity):** the share of actual defaults that the model catches. Our main goal was a high default recall, because every missed default is a loan made to a borrower who does not pay it back.
- **Precision:** the share of predicted defaults that really are defaults. Low precision means turning away many borrowers who would have repaid.
- **Specificity:** the share of non-defaults that the model correctly approves.

3.2 Logistic regression on the original data (LR1) Our first model, LR1, is a logistic regression trained on the training set with default status as the dependent variable. We used every other variable as a predictor except loan grade. Loan grade is the lender's own risk rating, so it is the result of a credit risk analysis that has already been done; we wanted to predict risk from the borrower's information instead. Interest rate has the same issue to a lesser degree, since lenders set it partly by grade (the average rate rises from 7.3% for grade A to 20.2% for grade G). We kept interest rate because it is known before the lending decision, but its strength as a predictor partly reflects the lender's grading.

All logistic regression models classify a borrower as a default when the predicted probability of default is above 0.5.

Fig 3.1: LR1 Confusion Matrix



Accuracy 84.2% | Default recall 48.9% | Precision 70.4% | Specificity 94.2%

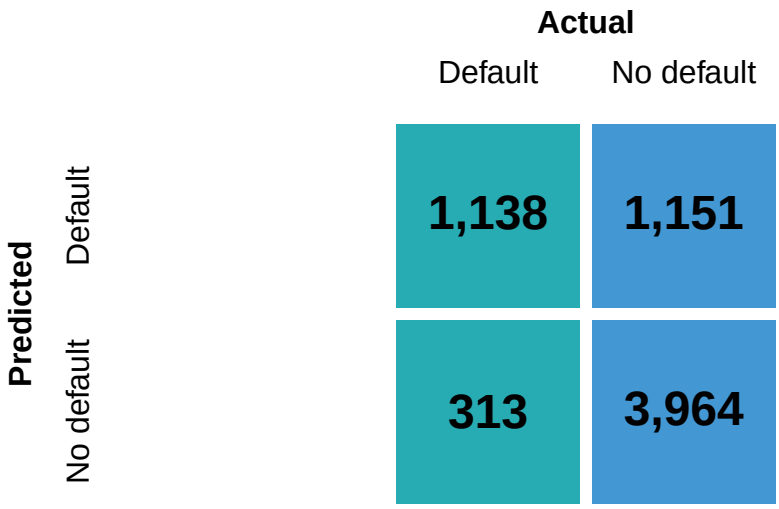
LR1 had an accuracy of 84.2%, but it caught only 48.9% of the defaults in the test set. The high accuracy comes mostly from correctly approving non-defaults, which make up most of the data.

3.3 Balancing the training data (LR2) One likely reason for LR1's low default recall is class imbalance: only 22.1% of the training set defaulted, so the model can reach a high accuracy while predicting few defaults. To correct this, we used undersampling, which randomly discards observations from the majority class until both classes are the same size.

We undersampled only the training set. The test set keeps the real default rate, so the results still show how the models would perform on new applicants.

The balanced training set has 8,708 observations, 4,354 in each class. We trained LR2 on it with the same predictors as LR1.

Fig 3.2: LR2 Confusion Matrix



Accuracy 77.7% | Default recall 78.4% | Precision 49.7% | Specificity 77.5%

LR2's default recall rose from 48.9% to 78.4%, while its accuracy fell from 84.2% to 77.7%. The cost shows up in precision, which fell from 70.4% to 49.7%: LR2 catches many more defaults, but about half of the borrowers it flags would have repaid.

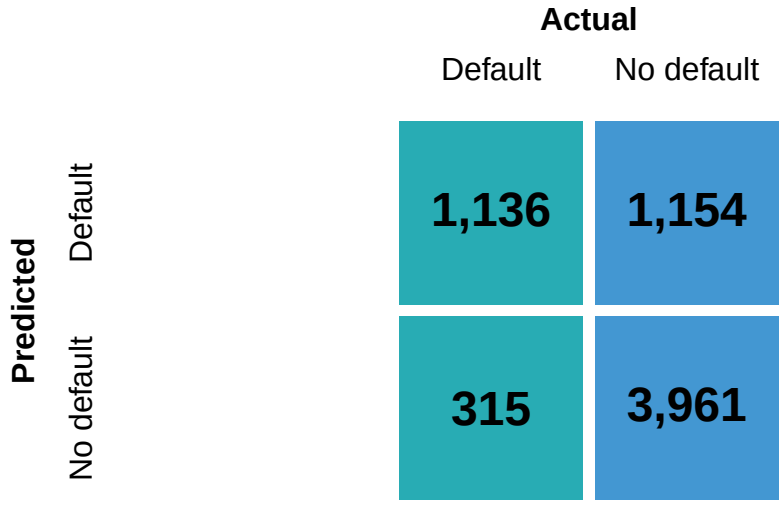
3.4 Removing insignificant predictors (LR3) Next, we tested whether each predictor in LR2 was significant using a likelihood-ratio test, which compares LR2 with the same model minus that predictor.

Table 1: Likelihood-ratio tests for each predictor in LR2

Predictor	p-value
person_age	0.718
person_income	0.007
person_home_ownership	< 0.001
person_emp_length	0.511
loan_intent	< 0.001
loan_amnt	< 0.001
loan_int_rate	< 0.001
loan_percent_income	< 0.001
cb_person_default_on_file	0.047
cb_person_cred_hist_length	0.437

person_age, person_emp_length, cb_person_cred_hist_length had p-values above 0.05. LR3 is LR2 without them.

Fig 3.3: LR3 Confusion Matrix



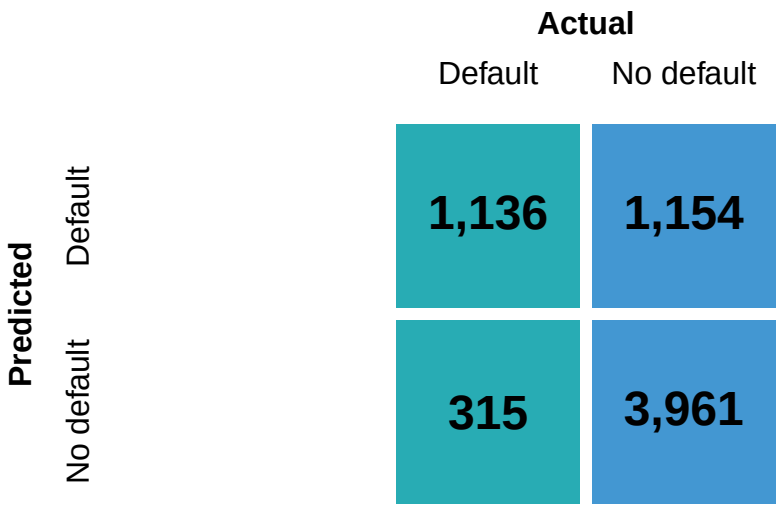
Accuracy 77.6% | Default recall 78.3% | Precision 49.6% | Specificity 77.4%

LR3 performed almost the same as LR2 (77.6% accuracy and 78.3% default recall) with 3 fewer predictors, so those predictors added little.

3.5 Stepwise selection (LR4) Instead of removing predictors by p-value, we also tried stepwise selection, which adds and removes predictors one at a time to minimize the AIC, starting from LR2.

Stepwise selection kept person_income, person_home_ownership, loan_intent, loan_amnt, loan_int_rate, loan_percent_income, cb_person_default_on_file. This is the same set of predictors as LR3, so the two selection methods agreed and LR4 is the same model as LR3.

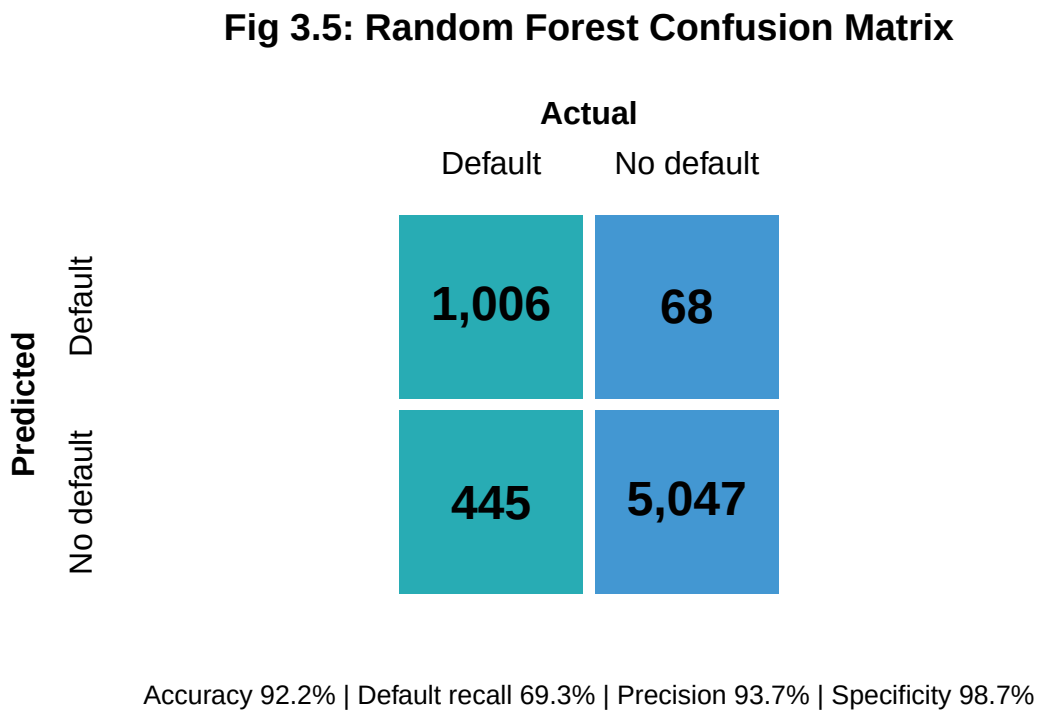
Fig 3.4: LR4 Confusion Matrix



Accuracy 77.6% | Default recall 78.3% | Precision 49.6% | Specificity 77.4%

LR4 had 77.6% accuracy and 78.3% default recall, practically the same as LR2 and LR3. Changing the predictors made little difference; balancing the training data was what changed the results.

3.6 Random forest Finally, we built a random forest of 500 trees. A random forest combines many decision trees, each trained on a random sample of the data and considering a random subset of the predictors at each split (here 3 of the 10). It can model interactions and non-linear effects that logistic regression cannot. We used the same predictors as LR1 and trained it on the full training set, without undersampling.



The random forest had 92.2% accuracy and 93.7% precision: when it predicts a default, it is almost always right. Its default recall of 69.3% is higher than LR1's, the other model trained on the same imbalanced data, but lower than the balanced logistic regression models.

Fig 3.6: Random Forest Variable Importance

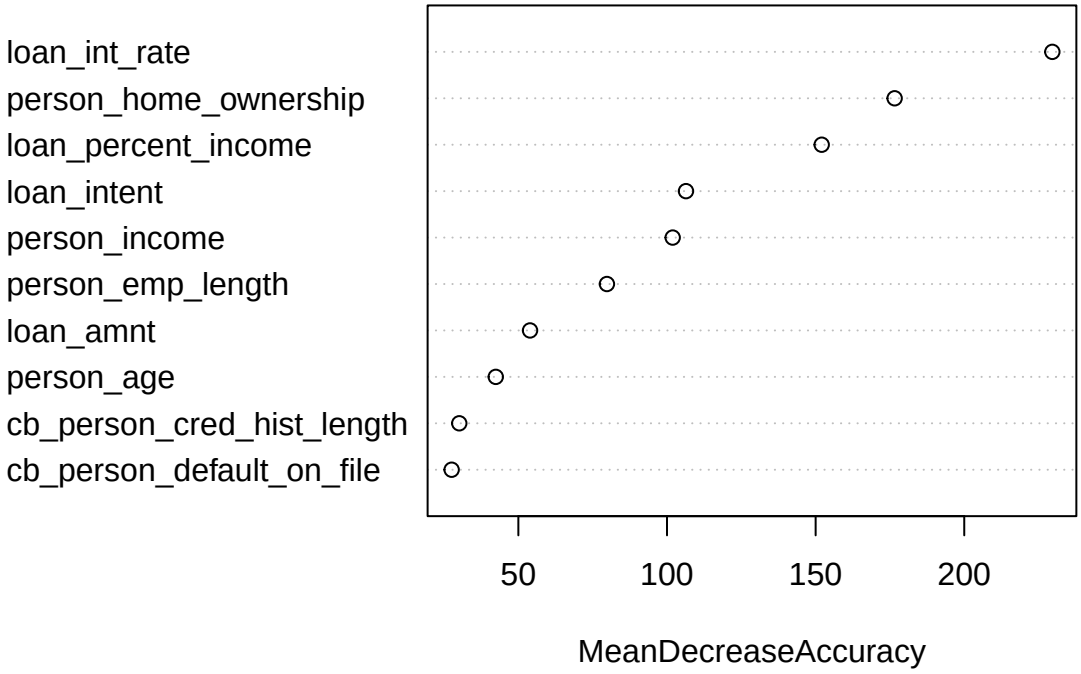


Figure 3.6 ranks the predictors by how much the random forest's accuracy drops when each one is randomly shuffled. Interest rate, home ownership, and loan-to-income ratio were the most important; all three also had p-values below 0.001 in LR2 (Table 1). Past default on file and credit history length were among the least important.

3.7 Model comparison

Table 2: All models on the same test set (positive class: default)

Model	Accuracy	Default recall	Precision	Specificity	Balanced accuracy
LR1	84.2%	48.9%	70.4%	94.2%	71.5%
LR2	77.7%	78.4%	49.7%	77.5%	78.0%
LR3	77.6%	78.3%	49.6%	77.4%	77.9%
LR4	77.6%	78.3%	49.6%	77.4%	77.9%
Random forest	92.2%	69.3%	93.7%	98.7%	84.0%

The balanced logistic regression models catch the most defaults, while the random forest is the most accurate and the most precise. Which model is best depends on the lender's costs: if a missed default costs much more than a lost customer, LR3 is the better choice; if both mistakes cost about the same, the random forest is.

4.0 Conclusions

Interest rate, loan-to-income ratio, and home ownership were the most important predictors in the random forest, and all three were highly significant in the logistic regression models. Loan intent also ranked highly in both.

Having a past default on file was a weak predictor once the other predictors were in the model: its p-value in LR2 was 0.047, and it was the least important variable in the random forest. This does not mean past defaults are unrelated to risk: borrowers with a past default paid an average interest rate of 14.5% compared with 10.3% for those without one, so interest rate already carries much of that information.

The largest improvement in default recall came from balancing the training data, not from changing predictors. Balancing also lowered precision, so it is a trade-off rather than a free gain.

Testing every model on the same untouched test set was essential. In the first version of this report, the random forest was tested on data that partly overlapped its training data, which made its accuracy look much higher than it is. Keeping the test set separate from every training and selection step gives results that reflect how the models would perform on new applicants.

Finally, the right model depends on how a lender weighs the two kinds of mistakes. The random forest is more accurate and precise, while the balanced logistic regression catches more defaults and is easier to explain.

5.0 References

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DataCamp, Fraud Detection in R, "Imbalanced class distributions": https://campus.datacamp.com/courses/fraud-detection-in-r/imbalanced-class-distributions?ex=5

"Surviving in a Random Forest with Imbalanced Datasets," SFU Professional Computer Science, Medium: https://medium.com/sfu-cspmp/surviving-in-a-random-forest-with-imbalanced-datasets-b98b963d52eb